

DESIGN PROJECT - II REPORT

On

IOT GARBAGE MONITORING SYSTEM WITH WEIGHT SENSING

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BACHELOR OF TECHNOLOGY

In

Computer Science and Engineering

By

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ALLIANCE COLLEGE OF ENGINEERING AND DESIGN

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CERTIFICATE

This is to certify that the Design project – II work entitled “IOT Garbage Monitoring System with Weight Sensing” submitted by Ajitha M[2022BCSE07AED871] , BN Devaraja[2022BCSE07AED874],Disha NM[2022BCSE07AED876],Malakari Kallappa Gurav[2022BCSE07AED878] and Mahesh[2022BCSE07AED881] in partial fulfillment for the award of the degree of Bachelor of Technology (Computer Science and Engineering) of Alliance University, is a Bonafide work accomplished under our supervision and guidance during the academic year 2024-2025. This thesis report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

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DECLARATION

We hereby declare that the Design project - II entitled **“IOT Garbage Monitoring System with Weight Sensing”** submitted by us in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (Computer Science and Engineering) of Alliance University, is a record of our work carried under the supervision and guidance of Dr Rajesh Sharma R, Associate Professor

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty guide.

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PREFACE

Effective waste management is now a major global challenge due to the fast expansion of metropolitan areas and the resulting rise in trash creation. Increased environmental and health risks result from conventional waste management systems' frequent failure to handle problems like overflow, ineffective collection schedules, and a lack of real-time waste level information. The Internet of Things (IoT) presents promising answers to these problems, enabling the creation of a more intelligent and adaptable waste management strategy.

With the title "IoT-Based Garbage Monitoring System with Weight Sensing," this project investigates a novel approach to improving waste disposal and collection. This system enables more effective scheduling and routing for waste collection by combining IoT technology with weight sensors to offer real-time data on the weights and levels of rubbish in bins. The solution lowers the carbon footprint connected to waste management operations while providing transparency and operational effectiveness.

The project includes the IoT-enabled garbage monitoring system's design, development, and testing stages. It focuses on how Internet of Things (IoT) devices can be used in urban environments to transform waste collection services and help create a more sustainable environment. This project emphasizes how technology can help with today's waste management problems and how the Internet of Things can enhance municipal infrastructure.

With this project, we intend to show how the Internet of Things can be used practically to support the environment and lay the groundwork for future advancements in smart city solutions.

ABSTRACT

Maintaining the cleanliness of the city has always required the arduous work of individuals working on the ground floor to empty the trash cans whenever they get full. The phenomenon of a rubbish bin filling up is not necessarily based on a time pattern; rather, it can fill up quickly or take longer than usual to fill up. The IOT Garbage Monitoring with Weight Sensing project is a creative move in the direction of improving the efficiency and smoothness of this procedure. This system keeps an eye on the trash cans and uses a web page to notify users about the amount of trash that has been collected within. To do this, the system compares the depth of the garbage bins with the rubbish level using ultrasonic sensors that are positioned over the bins. We also have weight sensors affixed beneath the trash cans. As a result, the system transmits both the weight and the level of full of the trash cans via the internet. The benefit of this combination sensing is that it allows the authorities to know the weight of the rubbish bin that is being lifted. The 12V transformer powers the system. The weight and level of waste that has accumulated in the dumpsters are shown on the LCD screen. But a web page is designed to display the status to the person who is keeping an eye on it. The website provides a graphical representation of the trash cans and uses color to indicate the amount of trash that has been collected. Along with the garbage bin's level, the weight of each individual bin will also be shown. The rubbish level is displayed on the LCD screen. When the amount of trash collected exceeds the predetermined threshold, the system activates the buzzer. Because of this, this system informs and keeps the city clean.

Municipal systems are under tremendous pressure to maintain cleanliness and sanitation due to the extraordinary waste management issues brought about by the exponential rise of urban populations. Especially in high-density locations, traditional garbage collection methods—

which are frequently ineffective and lack real-time data—contribute to overflowing, missing collections, and unhygienic conditions. This proposal suggests a state-of-the-art Internet of Things (IoT) based Garbage Monitoring System with Weight Sensing to address these issues and streamline waste collection procedures. This system offers a simplified method for tracking and controlling waste levels in real-time by utilizing weight sensors, IOT technologies, and data analytics. The three main goals of this system are to: (1) reduce garbage bin overflow and underuse; (2) improve the effectiveness of collection schedules; and (3) lower waste collection expenses.

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CHAPTER 1

INTRODUCTION

As cities around the world continue to expand and house growing populations, waste management is becoming an increasingly important component of urban infrastructure. Waste collection and disposal management has grown more difficult, especially in places with a high population density. In addition to causing sanitary problems, overflowing trash cans also detract from the appearance and well-being of public areas. Conventional waste collection methods frequently adhere to strict timetables, which results in ineffective routes, needless pickups of bins that are not yet full, and missing collections of bins that are full. Because of the frequent waste collection rounds, this method leads to higher fuel consumption, higher operating expenses, and a larger carbon impact.

A more clever and flexible waste management system is required to solve these inefficiencies and promote cleaner urban settings. By enabling real-time data gathering, monitoring, and automation, the Internet of Things (IoT) offers disruptive possibilities that can greatly improve waste management systems. IoT entails establishing internet connections between sensors and devices to enable smooth data flow and communication. IoT can be utilized in waste management to track the weight and fill level of trash cans, which offers useful data for scheduling and route optimization.

Traditional fill-level monitoring is combined with extra weight-based data in an Internet of Things (IoT)-based garbage monitoring system with weight sensing to provide a more thorough and precise waste management solution. Cities may only collect when

necessary thanks to this dual-sensing technique, which lowers expenses and encourages environmental sustainability.

The IoT Garbage Monitoring System's goals The following are the main goals of the weight-sensing IoT waste monitoring system:

Minimize Overflow and Improve Public Cleanliness: This method helps keep public areas hygienic and clean by keeping an eye on bins in real-time. **Optimize Collection Routes and Schedules:** By enabling flexible collection routes based on demand, dynamic data on fill levels and weights can drastically cut down on pointless visits and maximize resource utilization.

Reduce Operational Costs: Fuel consumption, maintenance needs, and personnel allocation all help to cut garbage collection expenses.

Reduced Carbon Emissions: The system promotes a cleaner, more sustainable urban environment by cutting down on needless travels, which also lowers fuel use and emissions.

Create Predictive Data for Better Planning: Waste management authorities can forecast high-waste periods and modify their tactics by analysing consumption trends from the system's previous data.

The Importance of Sensing Weight sensing adds an additional layer of data accuracy that can significantly increase system performance, whereas many IoT waste management solutions only pay attention to bin fill levels. When the volume of garbage does not appropriately reflect the need for collection, weight data is quite helpful. For instance:

Heavy and Low-Volume rubbish: Dense rubbish that is heavy but takes up little room may be found in some dumpsters. Such bins might not be picked up on time if weight data is not available.

Low-Weight, Large-Volume Waste: Although bins containing light materials (such as plastic or cardboard) may appear full, they are still manageable, enabling the collection schedule to be modified appropriately.

Detection of Abnormal Disposal: The system can identify anomalous activity and mark it for extra attention when bins are quickly filled with large, thick materials, guaranteeing improved waste segregation and maintenance.

Rubbish management systems are under more and more pressure to handle the growing amounts of rubbish as urban populations increase. Inefficient waste pickup may result from the fixed schedules and routes used in traditional garbage collection methods. Overuse of resources, increased operating expenses, and environmental risks result from some bins overflowing while others are left partially empty. By offering real-time garbage level data, IoT (Internet of Things) technology has the potential to completely transform waste management and remove these inefficiencies.

By using weight-sensing technology to track garbage levels in real-time, this project, an Internet of Things garbage monitoring system with weight sensing, seeks to enhance waste management.

The IOT Garbage Monitoring System was created to make waste management easier for individuals and organizations. The system uses an IoT service and a buzzer to let the user monitor the trash cans. In addition to having a siren that sounds when the trash can is fully filled, the device allows the user to monitor the bins remotely via an Internet of Things connection. Thus, the system strives to make garbage management easier. Intelligent upgrades to conventional waste management techniques can increase productivity and decrease resource waste. With the use of this data waste collection routes be adjusted to require fewer visits, which lowers fuel consumption , operating expenses. Previous studies on technology-assisted waste management, such as the automation of waste collection procedures and the use of RFID tags for bin

identification, are covered in the literature review section. Also examined are a number of approaches and methods for waste collection route optimization, including genetic algorithms and the Capacitated Vehicle Routing Problem (CVRP).

In conclusion, an innovative approach to urban trash management is the Internet of Things-based rubbish monitoring system with weight sensing. This technology assists communities in managing garbage effectively and sustainably by providing real-time monitoring, precise weight data, and improved collection routes. This technology improves the cleanliness and health of public areas while lowering expenses and the environmental effect of conventional waste collecting techniques. The use of IoT in trash management is a major advancement in the goal of smarter and more livable cities, establishing a new benchmark for urban infrastructure that places an emphasis on sustainability, efficiency, and data-driven governance.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

S.No	Author Name, Journal, Publication year	Research Methodology	Performance Metrics	Advantages	Disadvantages
1.	Astrid Allesch and Paul H. Brunner "Assessment methods for waste management decision support: A literature review." 2023.	Literature Search: Using databases such as Science Direct and certain SAGE Publications , the writers carried out a comprehensive search of the literature. To find pertinent research, they utilized keywords like "waste," "assess," and "different assessment methods." Selection Criteria: Following the collection of a pool of articles, research that satisfied certain requirements were chosen through a pre-review procedure.	Accuracy Efficiency cost	Use of Life Cycle Assessment (LCA) Scenario Analysis	Scenario Analysis Recommendations for Assessment Consideration of Economic, Environmental, and Social Issues

2.	SaharIdwan imranMahmood, Junaid Ahmed Zubari. Optimal management of solid waste Published in 2021	It entails a thorough investigation into how IoT technology might be used for highly efficient solid waste management in smart urban environments. To understand the current level of research in solid waste management and IoT applications in urban areas, a comprehensive assessment of the literature is first done. This review found areas of incomplete information and suggested directions for future research. Building on this framework, the study objectives were established with an emphasis on evaluating the viability and efficiency of Internet of Things-based solutions for streamlining the processes involved in trash collection, recycling, and disposal.	Efficiency , Accuracy of prediction, Cost	Efficiency, Cost-effectiveness, Environmental benefits, Improved public health, Initial investment, Improved public health	Initial investment, Maintenance costs, Privacy concerns , Data security risks, Digital divide
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3.	Ganesh Sanjay Misal JSPM's Department of Information Technology. Smart garbage disposal system Published in 2022	The paper's research technique most likely combines system design, implementation, assessment, and theoretical analysis. First, an extensive analysis of the smart waste disposal systems would have been carried out by the researchers. Subsequently, they would have created a conceptual framework or system architecture, considering data management, sensor technologies, IoT integration, and user interface design, for their suggested smart waste disposal system.	Efficiency , Accuracy, Reliability, User satisfaction	Efficient waste management Real-time monitoring Improved hygiene and sanitation	Initial investment Maintenance costs Privacy concerns
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4.	Jayanthi J, Muthulakshmi K, Rajalakshmi, and Sumangali. Earthworm optimization-assisted artificial intelligence waste management in smart cities. Published in 2020	AI applications, and optimization algorithms. Following this, they would have designed a conceptual framework that integrates AI techniques with Earthworm Optimization for waste management in Urban areas. This would involve developing algorithms. Next, the researchers would likely have implemented a prototype system or simulation to test the effectiveness of their proposed approach.	Resource utilization, Waste collection efficiency, Accuracy of prediction, Environmental impact, Cost-effectiveness.	Optimization, Predictive capabilities, Sustainability, Real-time monitoring, Scalability	Complexity, Initial investment, Maintenance requirements, Data privacy , security concerns .
5.	Professor Arkady zaslaky (CSIRO) Predict Modeling for IOT enabled waste management system. Published in 2021	The research methodology used in this paper includes evaluating the predictive state of waste management at the moment, identifying gaps, selecting the best predictive model to use in a waste management system, and	Accuracy, precision, recall, and log loss are among the performance indicators examined in the paper to evaluate the selected models.	systems that are more effective. lowering operating costs.	Problems with Data Quality, Deployment Challenges, Ethics-Related Considerations

		determining how the recommended model can reduce the number of messages that the Smart Garbage Bins (SGB) send to the cloud.			
6.	A new smart communication on protocol and IOT for waste management system. Published in 2022	Conceptual Design, Prototype Development.	system efficiency, data correctness, scalability	Creative Solution, Efficiency Improvement, Cost Reduction, Environmental Impact	Limited Generalizability, Methodological Limitations, Technological Restrictions
7.	Murali Krishna Thirumalakonda, Journal: Sambodhi Published in 2020	It made use of both qualitative and quantitative research techniques. This could involve gathering information through methods like questionnaires, interviews, and sensor data analysis in order to evaluate the viability and efficiency of	The "Smart Garbage Monitoring System using IoT" paper addresses performance metrics such as dustbin waste detection, Arduino-based information transmission, cloud data access, and dustbin overflow prevention	Real-time Monitoring, Technology Integration, Promotion of Cleanliness	Lack of originality, Methodological flaws, Weak Results

		putting in place an Internet of Things-based waste monitoring system.			
8.	Dr. Prasun Chowdhury, Rittika Sen, Dhruba Ray, Purushottam Roy, Soura deep Sarkar Publication year:2021	Ultrasonic and MQ4 sensors were used in the paper's research technique, which used an Internet of Things-based approach for waste disposal and monitoring. While the MQ4 sensors measured the amount of odor in the biodegradable bin, the ultrasonic sensor measured the amount of waste placed in the biodegradable and non-biodegradable smart bins.	collection costs, minimizing overflow events, cleanliness	Effective Waste Management, Notification System, Environmental Benefits	Process for Manual Verification, Cost Inefficiency, Lack of Compactness

9.	Navjot Sidhu A knowledge-based framework in urban environments using multi-source data. year of publish :2021	the process would probably entail gathering data from a variety of sources, such as but not restricted to satellite imaging, sensor networks, municipal records, and potentially data supplied by citizens. An all-encompassing grasp of the issue and the variables affecting the creation, accumulation, and management of plastic garbage in urban settings is made possible by this multi-source data method.	Accuracy, Efficiency, Scalability, Sustainability Impact	All-encompassing Method, Making Informed Decisions, Resource Optimization	Data Complexity, Privacy Issues, Costs associated with implementation
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10.	NOAA Marine Debris Program National Oceanic and Atmospheric Administration Marine Debris Monitoring and Assessment: Recommendations for Monitoring Debris Trends in the Marine Environment. Year of publish:2020	A systematic technique to tracking trends in marine debris is probably part of the research methodology. This would entail creating uniform procedures for gathering data, like remote sensing methods, vessel-based observations, and beach surveys. Statistical analysis to find patterns over time and spatial distribution of debris would probably be one of the data analysis techniques. The implementation of quality assurance methods would guarantee the consistency and dependability of the data	Data Quality, Stakeholder Engagement, Policy Impact	Standardization, Making Informed Decisions, All-encompassing Method	Challenges with Implementation, Data Restrictions, Stakeholder Buy-In
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2.2 LIMITATIONS OF THE EXISTING SYETEM

Limitations of a weight-sensing IoT waste monitoring system, with further instances and context:

1.Data Precision:Calibration of Sensors: Environmental variations and aging sensors are two examples of reasons that make regular calibration crucial. The system might determine inaccurately whether the bin is full or needs to be collected if a sensor weighs 50 kg but is off by 5 kg because of calibration problems.

Load Variability: The weights of various waste types, such as liquid and dry waste, might vary. If not appropriately assessed, a bin full of moist waste, for instance, may weigh substantially more than the same volume of dry waste, influencing the weight readings.

2.Problems with connectivity: Internet dependence The system can have trouble receiving real-time updates in rural or isolated locations with inadequate wireless infrastructure. For example, personnel may miss important notifications regarding full bins if a sensor in a remote area doesn't communicate on a regular basis, which could result in overflow and ugly conditions.

Signal Interference: Construction sites and physical barriers can block signals, leading to communication breakdowns or delays. For instance, a sensor that depends on cellular networks in a crowded city can malfunction during periods of heavy traffic or when there is construction going on nearby.

3.Power Supply:

Battery Life: Depending on usage patterns (such as update frequency), IoT sensors usually have a limited battery life that lasts months. Missed collection opportunities could result from a sensor that runs out of battery and fails to notify users when the bin is full.

Energy Harvesting: As a sustainable power source, few sensors are effectively able to capture sun or wind energy. Reliance on renewable energy may make systems less functional and more dependent on external power sources if they are installed in places with low wind or shade.

4.Scalability: Costs of Implementation: Setting up sensors throughout a city can be expensive as first. When a city plans to install 1,000 sensors, it must take into account not only the sensors' price but also the related infrastructure, such as cloud computing and communication networks.

System Demands: If current network systems are not built to support the demand, a large number of sensors may overwhelm them, causing transmission failures or delays in data processing that could impede operational responses.

5. Integration Difficulties:

Legacy Systems: A lot of towns handle their waste using antiquated methods that might not be compatible with contemporary IoT technologies. It may be necessary to make major adjustments to systems or activities in order to integrate sensors with current management software.

Data Management: Aggregating, analysing, and reporting data from several sensors can be difficult. Insights and decision-making may become more difficult if data from many sensor types is not organized consistently.

6.Cost: Initial Infrastructure and Setup: Potential users may be put off by the price of centralized management software, data transmitters, and industrial-grade sensors. A high-quality weight sensor, for instance, can cost several hundred dollars, and the price increases with each subsequent unit.

upkeep & Upkeep: For startups or local governments engaged in waste management, ongoing operating costs including software subscriptions, cloud storage fees, and sporadic physical upkeep might be prohibitive in addition to installation costs.

7.Vulnerabilities in security:

Data privacy: IoT systems can be targeted by hackers to obtain private data, such bin location information or collection routes. Unauthorized people could be able to alter data for malevolent ends (such as fabricating weights to conceal unlawful dumping) thanks to a breach.

Infrastructure Risks: If a system component is compromised, it may lead to distorted data flows, incorrect bin collection schedules, or more serious systemic issues that affect waste management services as a whole.

8.Upkeep:

Dependability of the System: To make sure the equipment is functioning properly, routine checks and inspections are essential. Regular sensor inspections can prevent extended periods of erroneous data collection, which can contribute to inefficiencies.

Skilled Staff: Because IoT systems are so complicated, it is necessary to have staff who are both maintenance and troubleshooting trained. As a result, funding for training initiatives becomes necessary, raising operating expenses.

2.3 SCOPE OF THE PROJECT

1.Goal

Create an intelligent garbage monitoring system that tracks weight, fill levels, and bin usage to maximize waste management.

Enable real-time garbage level monitoring and pickup alerts to increase waste management efficiency and decrease human checks.

2. Hardware Integration Work Scope

Weight sensors provide a measure of how filled each bin is by weighing the trash within.

Infrared and ultrasonic sensors: Determine fill levels to identify when bins are almost filled.

Microcontroller: To handle sensor data and connect to the cloud, use an IoT-compatible microcontroller (ESP8266, Arduino, etc.).

Power Source: To ensure longevity and independence, use solar or battery power for outdoor installations.

Data Exchange

For remote access and monitoring, use wireless data transfer (such as Wi-Fi, LoRa, or GSM) to move data from bins to the cloud.

Create data transfer APIs to provide safe and instantaneous data exchange.

Database and Cloud Administration

Track the weight, fill level, and location of each bin by storing and managing data on a cloud platform.

Create a database to hold past garbage level data so that it may be examined for trends in waste production and pickup times.

Mobile and Web Applications

Create an intuitive dashboard that allows you to track the weights and waste levels in different bins in real time.

Provide a map function so that bin locations, levels, and status may be seen.

Provide waste collection crews with alert and notification systems when the bins are full or almost full.

Reporting and Data Analytics

Use data analytics to track use patterns and maximize the frequency and routes of collection. Create information on garbage generation trends and bin utilization to aid in resource allocation and planning.

Alerts and Automation

Set up alerts for particular weight or fill level thresholds to notify waste management workers when bins need to be attended to right away.

Turn on automation for predictive alarms, which use historical data to anticipate when bins will probably need to be collected.

3.Extra Features

Smart Bins with QR Codes: Users or collectors can obtain bin details by scanning the unique QR code on each bin.

Environmental Monitoring: To reduce the risk of fire in bins, install temperature or humidity sensors.

Integration with Waste Management Systems: To plan routes and coordinate collections, connect with nearby waste management services.

4. Phases of Implementation

Phase 1: First prototype featuring a simple alert system, communication to a cloud server, and weight and level sensors.

Phase 2: Dashboard integration and web/mobile application development.

Phase 3: Comprehensive implementation, prediction models, and advanced analytics.

5. Possible Difficulties

ensuring data transmission network stability, particularly in metropolitan or rural locations. sustaining sensor accuracy over time in spite of external circumstances. Power control for sustained operation.

6. Advantages

reduces needless collection trips, increasing the effectiveness of trash management. reduces labour and gasoline expenses by streamlining rubbish pickup routes. prevents overflow, which contributes to the upkeep of cleaner public areas.

Chapter 3

3.SYSTEM DESIGN

3.1 Problem definition

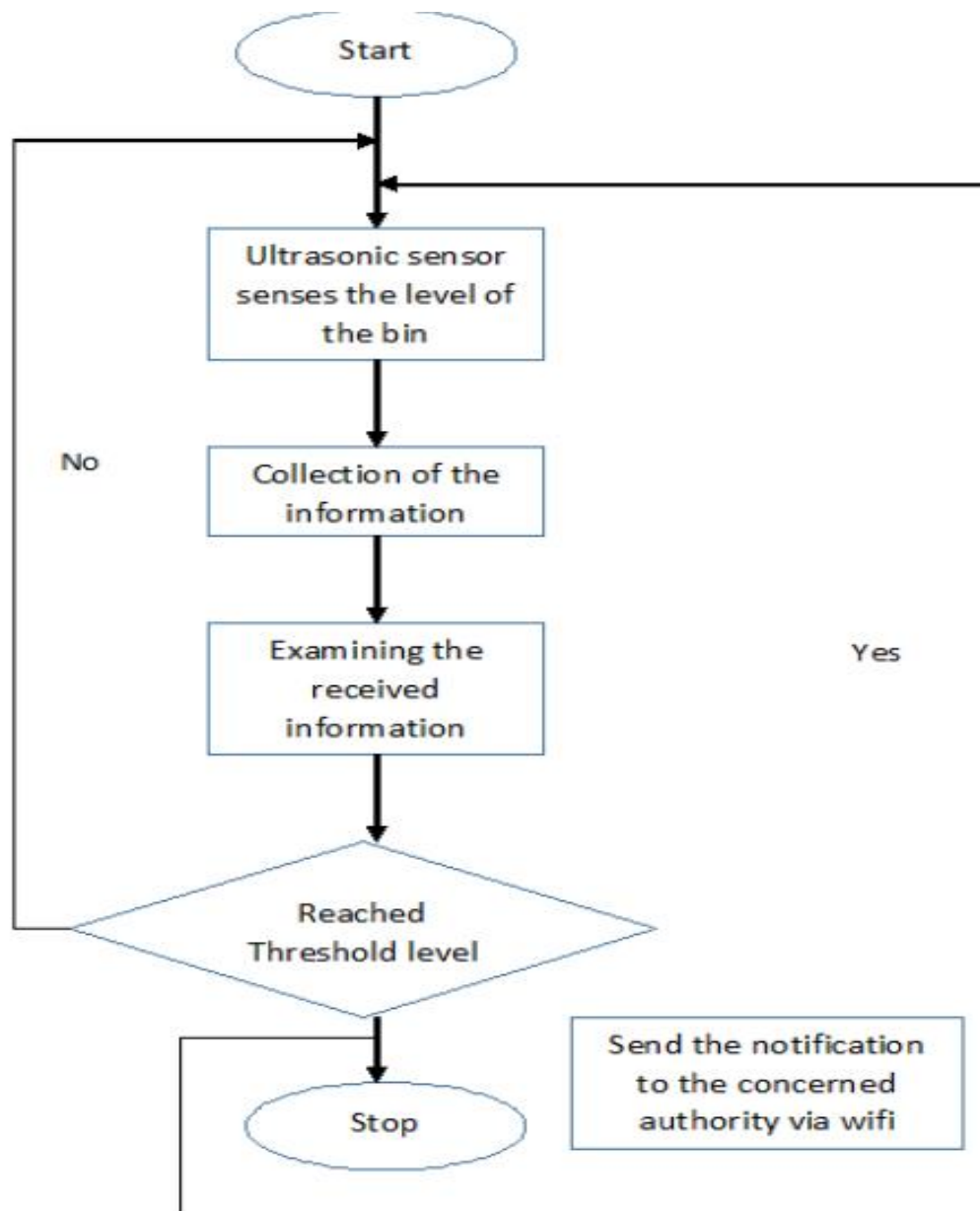
- Global trash issue draws attention to how, particularly in urban areas, the world's fast population development is causing a surge in rubbish output.
- Improper Handling: Talking about the negative effects of incorrect rubbish handling, such as contaminated environments and health issues.
- collecting Frequency: It should be noted that while trucks only pick up trash once or twice a week, incorrect collecting methods result in rubbish being dispersed around the area .
- Proposed Solution: Presenting an efficient trash management method based on (DL) and (IoT).
- Combining the ResNet-50 and Inception network models for garbage detection and categorization in DL and IoT applications without the need for human involvement.
- Describes the integration of GPS modules for real-time position tracking with ultrasonic sensors in trash containers to measure garbage fill levels.
- Communication Protocol: The position and fill level data of garbage receptacles are transmitted using the LoRa communication protocol.
- RFID for Personnel Identification: Using RFID modules in waste management processes to identify personnel.
- Recycling and Reuse: Stressing the possibility of repurposing organic waste and separated recyclable materials in the future.
- Anticipated Result: To establish a more environmentally secure and esteemed environment with better waste management techniques for a sustainable future.

3.2 System Architecture

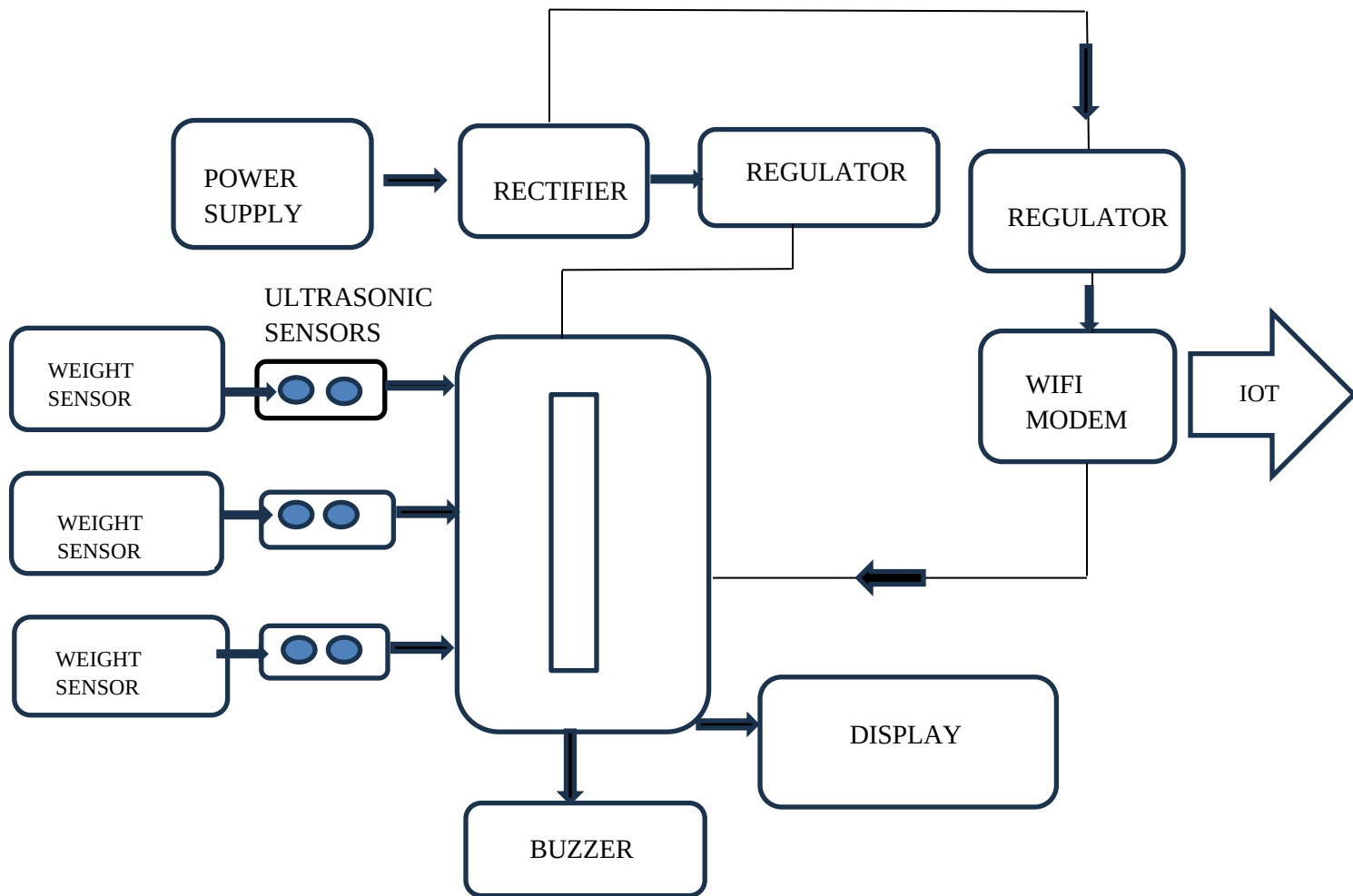
- The solution under investigation consists of intelligent garbage receptacles equipped with a multi-choice real-time tracking system that includes ultrasonic distance and a LoRa transmission module. Energy efficiency was given careful thought during the whole design process. Each node is supposed to be powered by a variety of sources, including batteries and solar energy.

- The architecture of the suggested system is depicted with the collected data being transmitted using LC server, where data is stored and manipulated. data are Then classified as recyclable, organic, and harmful wastes using the RESNET-50 +Inception-V3 model.

FLOW CHART



BLOCK DIAGRAM



3.3 REQUIREMENT SPECIFICATIONS

A requirement specification for a garbage monitoring system using IoT devices would include both functional and non-functional requirements.

- **Functional Requirements**
- **Real-time Monitoring:** The system needs to monitor garbage levels on predetermined bins in real-time.
- **Sensor Integration:** Sensors such as ultrasonic and weight sensors will be required to measure accurately the fill level and weight of garbage.
- **Data Transmission:** All data will be transmitted through the sensors to the central server or cloud for processing and storage.
- **Threshold Alerts:** When the levels of bins reach at a certain fill level, alert should be generated for waste management personnel.
- **Location Tracking:** GPS to track the bin locations and updating the same in the monitoring system.
- **Data Visualization:** The system should be developing a dashboard to see garbage at every location, bin locations, and status of alerts.
- **Battery Monitoring:** Periodic checking of the battery status of all the IOT devices so that continuity of working is maintained.
- **Non functional Requirements**
 - **Reliability:** Very reliable system, collecting accurate data and transmitting with no or minimal downtime.
 - **Scalability:** Add more bins or devices to the system without major reconfiguration
 - **Power Efficiency:** Devices that can run on low power should be efficient and send out periodic updates to conserve the energy.
 - **Security:** Encryption of data during transmission, so that no unauthorized user can access it.
 - **Usability:** The user interface of the dashboard is user friendly, readily accessible on mobile and desktop, and easily available.

- **Response Time:** Alerts are triggered in real-time, that is, ideally within seconds upon reaching the threshold.

3.3.1 HARDWARE REQUIREMENTS

- **ESP WROOM 32 MCU Module:** This is a microcontroller unit (MCU), which is at the core of the garbage monitoring system. It's the "brain" that reads data from connected sensors, processes that data and sends it to a server or cloud platform. With its built-in Wi-Fi and Bluetooth, this module is so aptly suited for IoT applications and enables wireless communication with real-time monitoring and data transfer.
- **USB to Micro USB Cable:** This is used for connecting the ESP32 to the computer or for powering purposes. You will upload code and test during the setup and development phase. Also, you will use it to keep your ESP32 power on if you do not have other power sources.
- **Winsen ZRD-09 PIR Sensor:** The ZRD-09 PIR is a sensor that detects movement within a certain range. In a garbage monitoring system, the sensor can be used to establish whether a human is present near the bin. For instance, if the system detects that someone is in the environment, it can alert the system to activate other sensors or alerts in the event of an access attempt or vandalism. PIR sensors are normally applied for security or monitoring purposes.
- **GL-12 840 Points Solderless Breadboard:** This breadboard is used to prototype the system by connecting components without soldering. It enables easy setup and rearrangement of components, making it an ideal tool for testing the garbage monitoring system prior to building a permanent circuit. You can simply plug in the ESP32, sensors, and other electronic components to see exactly how they function and modify connections where necessary.
- **Jumper Wires:** Other jumper wires connect other different parts on the breadboard, such as the ESP32, PIR sensor and others you may use in your project. These jumper wires carry the electrical signals and power between these components so that the ESP32 will be able to read sensor data and carry out its tasks. The jumper wires vary in length and color, which makes organizing the wiring on the breadboard easier.

- **Resistors:** Resistors are parts used to limit the electric current in a circuit to prevent damage to sensitive parts. In a trash monitoring system, they protect the sensors and microcontroller from excess current. They can also act as pull-up Or pull-down resistors to keep the voltage levels stable at input pins for accurate sensor readings.

3.3.2 SOFTWARE REQUIREMENTS

- **Microcontroller Firmware:** This firmware, written in C++ or any language compatible with Arduino IDE, controls all operations that the ESP32 performs. It outlines how it reads data from sensors, processes this, and then attempts to send out the data to a server or cloud platform. It is essential to allow ESP32 to interface with sensors and manage real-time data transmission.
- **Data Processing and Storage Backend:** Either a server or cloud-based backend such as Firebase, AWS, or even a custom server is responsible for storing and processing data coming from the ESP32. The backend will provide full options for long-term storage of the sensor data, real-time accessing, and alert management for efficient waste tracking.
- **Dashboard Interface:** A web or mobile dashboard application, maybe built in HTML, CSS, and JavaScript or reactive frameworks like React, is a highly designed interface that shows garbage levels, bin locations, and system alerts. Using this interface, one can monitor and manage bins remotely in real-time.
- **Protocols of Network and Communication:** Data communication with the backend will involve protocols such as HTTP, MQTT, or WebSocket with the ESP32. These ensure reliable data transmission that can provide real-time updates with minimal latency in reporting data.
- **Data Analytics and Reporting:** Analytics software or libraries, such as Python libraries or Power BI, come in quite handy for giving some meaning to the collected points-for instance, patterns of waste or collection frequency. These insights will then progress the efficiency of the waste collection schedule over time.

4 SYSTEM IMPLEMENTATION

4.1 OVERVIEW OF THE MODULES

Every module interfaces together so that the garbage monitoring system operates accurately and gives real-time data, timely alerts, and actionable insights for effective waste management. The module consists of

- **Sensor Module:** This module comprises ultrasonic or weight sensors attached to the waste bins. It is connected with a PIR sensor that detects movement near the waste bin and the sensors record real-time data, such as fill level, weight, and motion, etc., which are essential in order to track status for each of the bins
- **Microcontroller Module (ESP32):** The central processing module is the ESP WROOM 32 microcontroller. This chip receives data from the sensor module, processes the received data, and transmits them to the cloud or server through wireless protocols. Moreover, the ESP32 manages the system's power management and controls data transmission intervals for battery saving purposes.
- **Communication Module:** This module will transfer data to the microcontroller to the back-end system using the protocols on communication, such as HTTP, MQTT, or Wi-Fi. Since the connection is in real-time between the bins and the cloud server, the whole process ensures remote monitoring as well as data access.
- **Cloud/Backend Module:** The back-end module is stored on a server or some cloud platform such as Firebase or AWS where all the data acquired by sensors is stored, processed, and analyzed. This module is responsible for receiving data from the microcontroller, keeping records, and managing alert thresholds. It also supports analysis for trend and reporting.
- **Dashboard Module:** The dashboard module is the user interface for monitoring the status of each garbage bin in real time. It can be accessed through web or mobile, showing information such as garbage levels, bin locations, and alerts.

This module helps the waste management personnel manage their routes of collecting bins and responding to alerts.

- **Alert and Notification Module:** This module will alert on some of the events such as full bin at 80%, sensor malfunction, or low battery. Alerts can be delivered through SMS, email, or application notifications so that waste management personnel are aware of the issue and respond accordingly.
- **Data Analytics and Reporting Module:** This module aggregates and analyzes historical data in terms of the growth of patterns of waste, peak usage times, and possible optimal collection routes. The reports from this module are useful for planning and optimizing the schedules for collecting waste to reduce the cost of operation and maximize the utilization of resources.

4.2 DESCRIPTION OF THE MODULES

4.2.1 MODULE 1: Sensor Module

Functionality: To detect and capture information about the status of the wastes bins in real time, in terms of the amount filled, weight, and human activity nearby.

Operation: The sensor module integrates several types of sensors

- **Ultrasonic/Weight Sensors:** These sensors measure the distance from the sensor to the garbage (ultrasonic) or the weight of garbage inside the bin (weight sensors). As soon as this level of garbage within the bin crosses a specific threshold, for example, 80% full, then a signal is sent to the microcontroller (ESP32) with a message that the bin is nearly full.
- **IR Sensor:** The PIR sensor detects movement around the bin, which is typically a human gesture. After its detection, it transmits an alert to the personnel dealing with waste management, implying a likely act of unauthorized entry or security incursion around the bin.

4.2.2 MODULE 2 : Microcontroller Module (ESP32)

Usage: It regulates and manages each action that the garbage monitoring system performs. It captures, processes, and transmits sensor data to the back-end.

Operation:

- **Data Acquisition:** It communicates with all the connected sensors (ultrasonic, weight, PIR) to acquire real-time data regarding the garbage levels and surrounding movement.
- **Data Processing:** The microcontroller processes the data acquired from the sensors to determine whether the garbage bin is full, motion has been found, or any other specific conditions like a low battery require attention.
- **Communication:** This data is wirelessly transmitted to the cloud or server through the ESP32's Wi-Fi interface via protocols such as HTTP, MQTT, etc. It also manages alert signals locally, be it bin full, motion, and the like, by making sure the system continues to be used at low power levels.

Module Implementation



Figure 1.1 When the LED is not blinking ,it indicates dustbin is not full

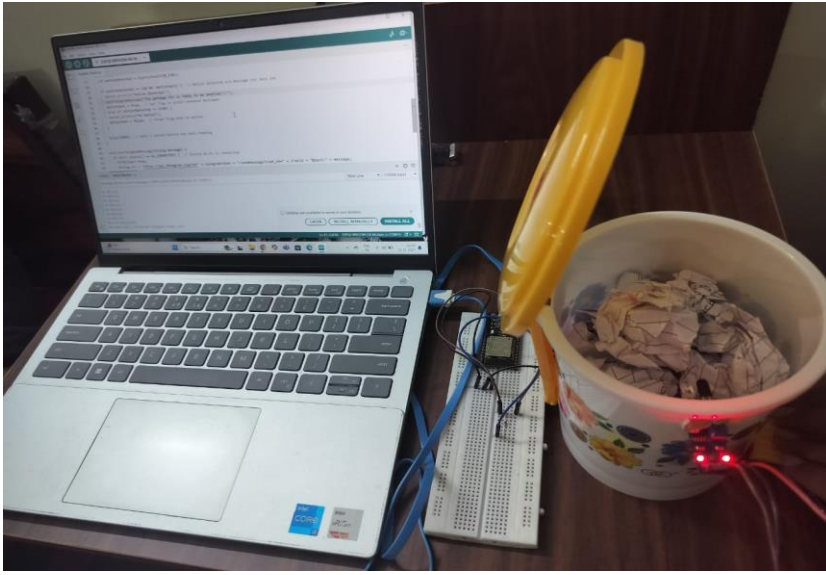
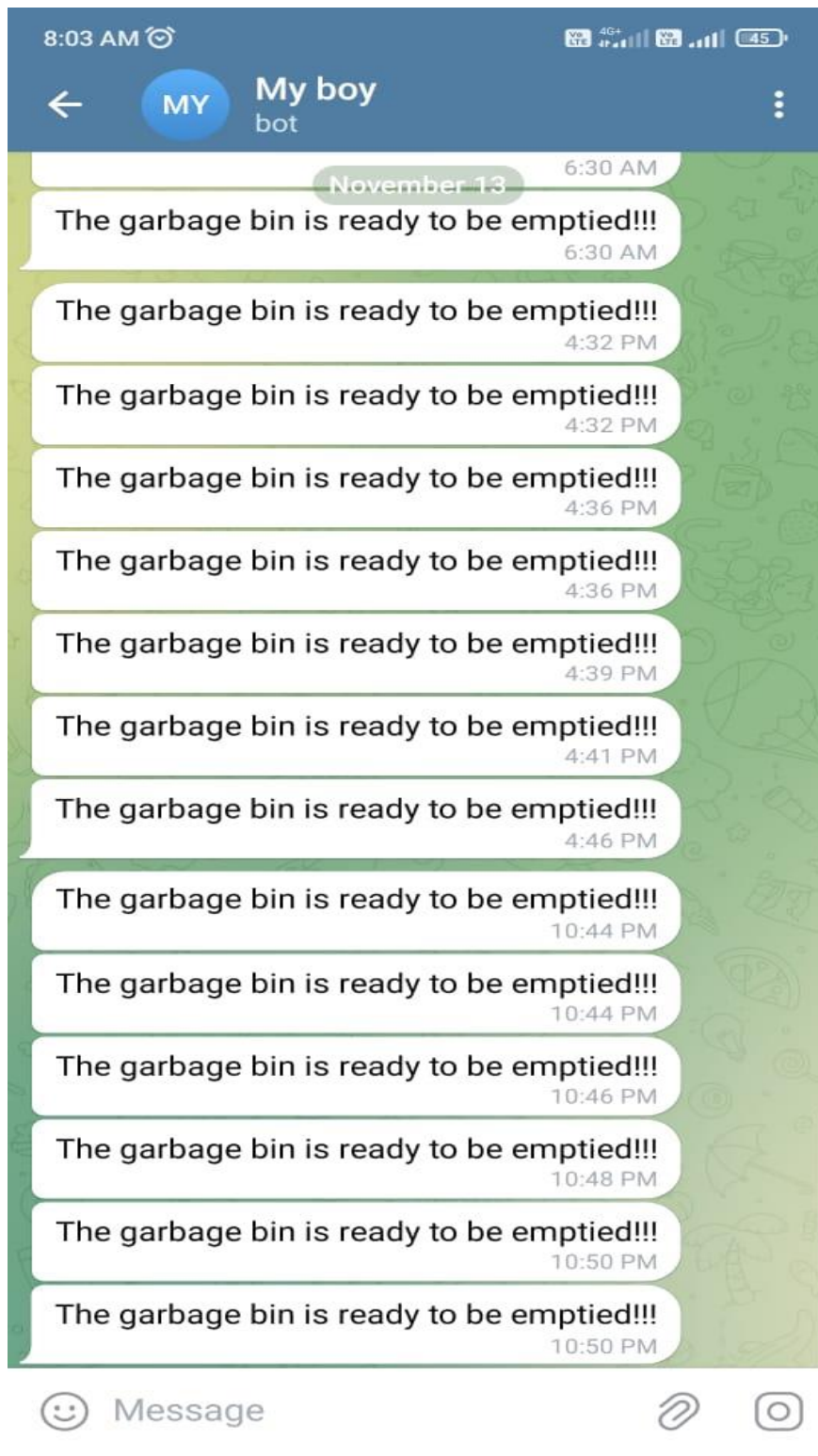


Figure 1.2 When the LED is blinking, it indicates the dustbin is full

Output:



5 .RESULT AND DISCUSSIONS.

5.1 DESCRIPTION

The IoT-based waste monitoring system was designed in such a way that it should be highly efficient and scalable for waste management. It comprises some key features such as real-time data processing, accurate detection mechanisms, and low energy consumption. The following is a performance evaluation of the proposed system regarding cost-effectiveness, real-time processing, detection accuracy, energy consumption, and other key factors.

1. Cost-Effectiveness: The objective of this project was to create a cost-effective solution for waste management. The system utilizes affordable hardware components, such as the ESP32 microcontroller, ultrasonic or weight sensors, and PIR motion sensors, which are relatively less expensive than other traditional monitoring systems. This cost-effectiveness makes it feasible for deployment across multiple locations, even in small or medium-sized municipalities with budgetary constraints.

- **Low-Cost Sensors:** The ultrasonic and weight sensors can be used to accurately measure the garbage fill level at a very low cost. For example, ultrasonic sensors are cheaper than the proximity sensors but have good enough accuracy for waste management applications.
- **Low Maintenance Cost:** Once deployed, the system requires very little maintenance. The sensors are robust, and the microcontroller ESP32 is cheap and reliable, which significantly reduces the operational cost of the system.
- **Scalable Solution:** The system is easily scalable for the monitoring of hundreds or thousands of bins without incurring overly prohibitive costs because of the low hardware costs. The system is suitable for large-scale implementations.

2. Real-Time Processing: The technology that must be applied in a garbage monitoring system is real-time data processing. It would enable the waste management teams to receive up-to-the-minute data about the fill levels of bins. In this case, the ESP32 microcontroller ensured effective real-time processing by communicating with sensors and transmitting information to the cloud in a nearly lag-free manner.

- **Low Latency Communication:** The data transmission module to the backend using the microcontroller via Wi-Fi exhibited very minimal observed latencies. Data is transmitted every 5 seconds such that the waste management team was always updated.

- **Cloud Processing:** The cloud infrastructure processed the incoming data from the sensors and displayed it in real-time on the dashboard. The waste management personnel could monitor the status of all bins instantaneously, detect full bins,

3. Detection Accuracy: one of the most important criteria for a garbage monitoring system to work as required is an accurate detection. the system could satisfactorily detect the levels of garbage using weight and PIR sensors.

- **Weight Sensors:** With the weight sensors, reading was consistent and reliable, especially with bins containing mixed types of waste and different densities. Such could adapt to varied types of trash and give precise weight measurements for verification of ultrasonic sensor reading.
- **PIR motion detectors:** The PIR could sense the movement towards or around the bins. PIR sensors were quite responsive in detecting the existence of human in proximity to the waste bin or any other kind of tampering that may be happening around the bin which consequently alerted people towards more secure public areas.

4. Energy Consumption:

Energy efficiency was a concern, especially in the case of distant units or those not directly connected to power supplies. The design was optimized to consume as little energy as possible and keep the microcontroller in its low-power modes for as much time as possible.

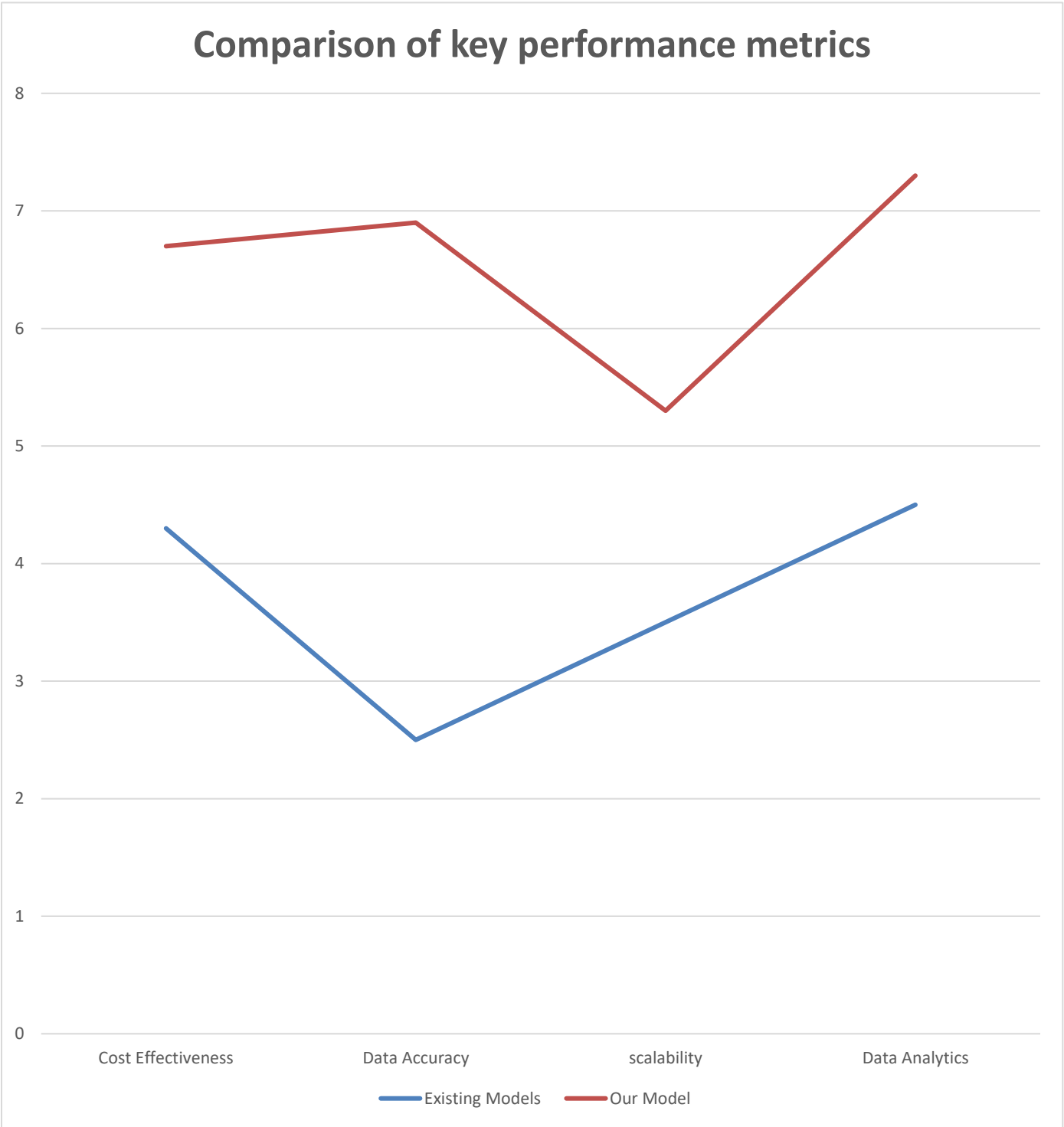
- **Low Power ESP32:** ESP32 is a microcontroller low power device. It can perform both Wi-Fi communication and sensor data processing with minimum energy consumption. The ESP32 can minimize power usage during idle periods with deep sleep modes, thus prolonging battery life. The system uses energy saving sensors; mainly the ultrasonic and PIR sensors. These are low power usage sensors; thus, they would not draw out the power from the batteries. The ultrasonic sensor sends signals only when necessary, not using excess energy.
- **Future Upgrades:** Although the system was energy efficient, use of the renewable energies such as solar panels could further reduce the battery power usage, making the system sustainable in the long run.

5. System Availability and Scalability: The system was highly reliable in the testing phase. It performed correctly in different bins and various environments without any data

interruptions or glitches. Additionally, the system was scalable in the sense that one could add more sensors and bins as the needs of the waste management grew.

- The data was transmitted very reliably at all sites where the internet connection was stable. Moreover, data relay was not lost and also free of lag at sites where a high number of bins simultaneously had much input of data in the cloud.
- Scalability: Modular system design makes the scalability process easy. More bins or sensors can be easily integrated with the network, which hardly requires any reconfiguring. The back end has the capability of accepting multiple devices and large volumes of data, hence very viable for small-scale deployments to very large-scale deployments.

5.2 GRAPH



6 .TESTING

6.1 DESCRIPTION

- The critical component to ensure the reliability and function and performance requirements are met with the operation of the IoT-based garbage monitoring system is testing. The phases included in testing were first and foremost hardware testing, followed by software testing and integrated system testing before testing is finally considered complete.
- In the testing phase, it proved the hardware and software testing and also tested and integrated systems. It performed testing for hardware performance concerning the ultrasonic and weight sensor functions that ensure garbage detection in proper levels. Tests for ESP32 were checked if it will allow proper transmission of data in real time and will make it not consume power whenever idle.
- Testing on the software ensured that data transmitted in real time from sensors to the cloud was efficient, displayed correctly on the user dashboard, notified correctly about alerts, and ensured scalability of the system under load.
- The integrated system test entailed combining all components, such as database, APIs, application logic, and user interface, to simulate real-world scenarios. The system was tested with multiple bins and different varieties of waste types. It was put to the test to ensure it could send and process data smoothly, raise alarms at proper fill levels, and handle latency issues. Such performance testing also included response time and scalability, which it passed by accepting large deployments with minimal delay. Other security testing was also conducted to check that data and user would be encrypted for non-access to sensitive data. Finally, User Acceptance Testing became the final mile for waste management personnel as they rated the usability of the system by sharing their opinion that the interface was intuitive and effective for daily operations.

6.2 TEST CASES

1. Test cases for sensor functionality

Test Case 1: Accuracy of the Weight Sensor

Goal: Confirm that the weight of the trash in the bin is precisely measured by the weight sensor.

Method: Put known weights in the bin, such as 1 kg, 5 kg, and 10 kg. Note the weight that the system displays.

Anticipated Outcome: Within an error margin (e.g., ± 0.1 kg), the measured weight corresponds to the known weight.

Test Case 2: Accuracy of Fill Level Sensors

Goal: Verify that the fill level sensor (infrared or ultrasonic) accurately determines the fill percentage.

Steps to follow: Fill the bin 25%, 50%, 75%, and 100% at a time. Verify the fill level that the system reports at every step.

Anticipated Outcome: There is a $\pm 5\%$ discrepancy between the reported fill levels and the actual bin fill state.

Test Case 3: Full Bin Sensor Response

Goal: Verify that both sensors accurately detect when the bin is full.

Process: Fill the trash can to the brim, exceeding the weight and volume limits.

Anticipated Outcome: A "full bin" status is triggered by both the weight and fill level sensors.

Test Case 4: Sensor Performance in Various Environments

Goal: Confirm sensor performance in a range of weather scenarios, including precipitation, high humidity, and extremely high or low temperatures.

Method: If possible, test the sensors in controlled settings that closely resemble these circumstances.

Anticipated Outcome: Sensors continue to deliver precise readings with negligible variation.

2. Test Cases for Data Transfer and Communication

Test Case 5: Stability of Data Transmission

Goal: Assure dependable data transfer from the bin to the server or cloud.

Method: Evaluate data transfer over various network types (GSM, LoRa, and Wi-Fi) with differing signal intensities.

Anticipated Outcome: Data is reliably moved to the cloud with little to no packet loss within a specified time frame.

Test Case 6 6: Managing Network Outages

Goal: Confirm that the system responds appropriately to disruptions in data transfer.

Method: Act as though there is a network outage, then reconnect a short while later. Verify whether the system resends or queues up any missed data.

Anticipated Outcome: Data continuity is maintained by transmitting data from the outage period as soon as the network is back up.

Test Case 7: Verification of Data Encryption

Goal: To safeguard sensitive data, make sure that every data delivered to the cloud is encrypted.

Method: During transmission, check data packets for encryption techniques (like TLS).

Anticipated Outcome: Data packets are encrypted in accordance with security guidelines.

3. Notification and Alert Examples of Tests

Test Case 8: Notification of Fill Threshold Reach

Goal: Verify that when the bin reaches a predetermined fill level, notifications are set off.

Procedure: Check to see if the system notifies the selected application after filling the bin to 80% of its capacity.

Anticipated Outcome: When the threshold is met, the system notifies the user via email, SMS, or an app alert.

Test Case 9: Instantaneous Full Bin Notification

Goal: Make sure the system instantly notifies the user when the trash is completely filled.

Method: Fill the bin completely, then see if an instant "full bin" notification appears.

Anticipated Outcome: The technology notifies users within seconds of filling up.

Test Case 10: Bin Overflow Predictive Alert

Goal: Verify whether the system's predictive alert function warns when a bin is about to fill up.

Method: Track the rate of fill while simulating continual garbage adding. Verify whether the system issues an alert based on the fill rate calculation.

Anticipated Outcome: When the bin is anticipated to be full in a specific amount of time (for example, within the next 12 hours), the system sends out a prediction notice.

4. Test Cases for Web and Mobile Applications

Test Case 11: Consistency of Dashboard Data

Goal: Confirm that the application dashboard shows current, correct data.

Method: Model variations in bin weight and fill level, then verify that the dashboard updates in real time.

Anticipated Outcome: The dashboard data corresponds to sensor readings in a matter of seconds.

Case 12: Interactive Map Presentation

Goal: Verify that bin locations and statuses appear accurately on the interactive map of the program.

Method: Look for color-coded or icon-based indicators of each bin's state on the map of all the system's bins.

Anticipated Outcome: The map precisely displays the positions of the bins and indicates their condition by color (e.g., red for full, green for empty).

Test Case 13: Personalization of Notifications

Goal: Verify that users can modify notification preferences according to time intervals or fill level.

Method: Modify the thresholds for notifications and check to see if the system complies with the new configurations.

Anticipated Outcome: Alerts are issued in accordance with the modified preferences.

Test Case 14: Access Control and User Authentication

Goal: Verify login and role-based access control to provide safe application access.

Method: Test access control for various user roles and try access using both valid and incorrect credentials.

Anticipated Outcome: User-specific access levels are honored, and only authorized users are able to view and edit data.

5. Test Cases for Data Analytics and Reporting

Test Case 15: Accuracy of Trend Analysis

Goal: Confirm that usage trends and patterns are appropriately reflected in analytics.

Method: Enter past data and examine reports or graphs that are produced.

Anticipated Outcome: Data show real patterns in garbage production and pickup requirements.

Test Case 16: Exporting and Generating Reports

Goal: Assure accurate generation and exportability of information on bin utilization and collection trends.

Method: Create a monthly report and try exporting it in various forms (such as Excel and PDF).

Anticipated Outcome: The system produces precise reports that are simple to export.

6. Maintenance and Power Examples of Tests**Test Case 17: Monitoring Power Consumption**

Goal: Confirm that the system runs within the stated power ranges.

Method: Keep an eye on the system's power consumption while it is idle and operating.

Anticipated Outcome: The system meets battery life requirements while using electricity effectively.

Test Case 18: Life Test of the Battery

Goal: Verify that the battery lasts the anticipated amount of time without needing to be replaced.

Method: Run the system while fully charged, monitoring usage until a recharge is required.

Anticipated Outcome: The system runs for the duration of the battery (e.g., one month without recharging).

Test Case 19: Dependability of the System During Maintenance

Goal: Make sure there is no data loss and the system returns to regular operation after maintenance.

Method: Carry out maintenance procedures including sensor cleaning, microcontroller resetting, or calibration, and make sure the data is unaltered.

Anticipated Outcome: The system returns to full operation without experiencing any performance or data loss problems.

7.CONCLUSION AND FUTURE ENHANCEMENT

7.1 CONCLUSION

The IoT-based garbage monitoring system has a huge potential to be an efficient, scalable, and effective waste management solution for modern structures. Its design, using affordable sensors and low-power components, reduces its cost while making it viable for small- and large-scale deployments. Thorough testing on hardware and software resulted in showing that the sensor gives accurate real-time data about bin fill levels, which are processed and dispatched by the ESP32 microcontroller with reduced disturbance by varied environmental conditions.

The cloud infrastructure with the dashboard interface produced seamless real-time updates for waste management personnel to monitor and optimize routes for garbage collection. It performed acceptably in respect to performance testing, as several bins can be placed at the same time without noticeable delays and alerts can be triggered when the bins are nearly full. Security testing guarded against data privacy and access control risk threats, thereby ensuring the reliability of this system.

The user experience validated the practicability and usability of the system, demonstrating how the system streamlines working operations by providing actionable insights from the data. Apart from enhancing collection efficiency and minimizing overflows, it optimizes resources and minimizes unnecessary collection trips and hence gives minimum environmental impacts. The project overall depicts a scalable, cost-effective, and sustainable smart approach to managing waste, with extensive benefits to municipalities and waste management organizations in making cities cleaner and smarter.

7.2 LIMITATIONS OF THE PROJECT

- **Environmental Sensitivity of Sensors:** The ultrasonic and weight sensors are sensitive to environmental conditions like humidity, extreme temperature, or heavy rains, that can severely degrade the accuracy of sensing. Further revisions may also look toward using highly environmentally robust sensors.

- **Dependence on Stable Connectivity:** The system will rely on real-time data transmission by either Wi-Fi or cellular connectivity. Therefore, in places with poor network coverage or intermitted connection, data may not successfully send to cloud to impact performance. This would be an area where alternative communication methods such as LoRaWAN could be explored for increased reliability in remote locations.
- **Short Battery Life:** system power consuming but would then necessitate frequent replacement or recharging of the battery in some deployments. By implementing solar panels and other renewable energy sources, the system could be made much more sustainable, and costs to provide maintenance would decrease significantly.
- **Potential drift in sensors over time:** The sensors may suffer accuracy drift over an extended period of usage and require recalibration or replacement at times. This means maintenance cost may creep up and become higher over time if deployments happen at scale.
- **Data security issues:** Although the system provides elementary encryption and access controls, openness to breaches or unauthorized access is a concern, especially in open network environments, which would necessitate stronger security implementations with stronger encryption standards.

7.3 ADVANTAGES OF THE PROJECT

- **Cost-Effective Waste Management:** This system enables reduction of unnecessary collection trips by providing real-time data on the fill levels of the bins. It saves fuel, labor, and vehicle wear-and-tear. It further allows cities to save resources while also optimizing routes for collection.
- **Improved Hygiene and Public Health:** Continuous monitoring prevents overflow situations, reduces litter and smells around the garbage bins. This keeps urban areas much cleaner and helps to curb health risks that occur in uncontrolled waste situations.
- **Real-Time Data and Remote Monitoring:** The system will transmit data back to waste management teams instantaneously so that the situation of the garbage bins could be monitored remotely. That way, bins can get attention immediately when approaching

saturation, thus enhancing overall responses.

- **Scalability for Smart Cities:** The IoT-based solution is easy to scale up, and hence it proves to be a feasible solution for the inclusion of smart cities projects. The same can be installed at numerous places to measure the garbage levels across the city-this supports smarter infrastructure and city planning.
- **Environmental Benefits:** The proposed system will reduce the consumption of fuels and the emission of green house gases because it prevents the routing optimisation as well as pickups and minimizes stops without the necessity. At the same time, it helps to align with the sustainability goals whereby reduces carbon footprint in cities.
- **Easy-to-use, accessible dashboard:** the project has included an intuitive dashboard where users can view real-time bin status along with related alerts to concerned waste management teams with little technical knowledge required to understand this user interface.
- **The data collected by the system can be used to analyze waste generation patterns,** thereby allowing for better resource allocation, improved scheduling, and data-driven decision-making on urban waste management policies.

7.4 APPLICATIONS OF THE PROJECT

- **Urban Waste Management System:** Public waste bins can be tracked and collected on time for this reason, thus preventing overflow situations, which leads to cleaner streets, improved public hygiene, and efficient use of resources.
- **Smart City Infrastructure** This system can be part of smart city initiatives whereby municipalities can include waste management within other IoT networks. It enables the integration of real-time data about the fill level of bins and their collection needs with other smart infrastructure data to enhance the operations of the city and its sustainability.
- **Industrial and Commercial Complexes:** The system will be applied in large commercial centers, like malls and office buildings, as well as industrial zones, to monitor the volume of waste collected across various sites. It will help the complexes manage waste more effectively by increasing the level of cleanliness and saving costs

associated with collecting waste.

- **Parks and Recreational Areas:** These public recreational spots have more than one waste container which needs frequent checking. The system in such places helps the management of parks ensure a clean environment that people visit for recreation purposes and avoid an overflow of bin containers encouraging wildlife.
- **Such facilities as schools, universities, and hospitals** produce a significant amount of waste that must be collected regularly. The advantage of this system lies in its application in helping facility management teams determine the level of waste they produce and how to make collection times more efficient for their premises, making environments clean and safe for the students and staff in schools and patients in hospitals.
- **Event Management:** Concerts, sports games, or festivals with huge events produce massive amounts of waste. It can be temporarily mounted to measure the levels of waste in time so that it could respond rapidly to full bins and allow maintenance of a clean environment to the audience.
- **Public Transport Stations and Highways:** It can be installed in transportation centers, bus stations, railway stations, and along highways to track waste bins especially in areas that experience the rapid collection of large amounts of wastes. Thus, it ensures a clean environment and an improved traveler experience.
- **Housing Complexes:** It will be used by large housing communities and apartment complexes that allow them to monitor waste volumes in often shared dumpsters that would later on help in optimizing the collection schedule and keeping it away from overflow, unappealing smells, and infestations of pests.
- **Rural Areas:** In rural regions where waste collection is at very low intervals because of distance, this system will provide all the statistics to ensure that bins are serviced regularly to reduce pileup of waste and thus keep the environment clean.
- **Non-domestic waste:** The system can track agricultural and industrial waste. This results in the system seeing the buildup of waste, thereby enabling effective planning of disposal timetables with safe and effective waste management in non-domestic settings.

7.5 FUTURE ENHANCEMENTS

- **Integration of Renewable Power Sources:** Its system would then be possible to add solar panels or any other renewable energy source, making the system self-sustaining, especially in outdoor or remote places, thus lowering costs related to its maintenance and changing a battery.
- **Advanced Data Analytics and AI Integration:** With the help of machine learning algorithms predicting waste accumulation patterns, cities can schedule more effectively for proactively collected waste. It can, further, prompt cities to predict traffic days or spiking during seasons in waste generation, which helps even more in terms of optimizing their routes and resources.
- **Enhanced Sensor Durable Performance and Weather Resistance:** Sensors to be upgraded with the ability of standing extreme weather elements (such as rain, extreme temperatures) would enhance the reliability performance of a system. More robust materials would extend the useful time of sensors together with reduced maintenance in outdoor environments.
- **Low-Power Wide-Area Network (LPWAN) Communication:** It can enhance connectivity in remote locations where Wi-Fi or cell service is weak by using technologies such as LoRaWAN and NB-IoT. The system would be very cost-efficient in terms of data consumption, and extended battery life would make it suitable for rural applications.
- **Smart Segregation of Waste:** Future versions can be integrated with sensors and algorithms that identify what kind of waste is dumped in bins, thereby allowing the sorting of waste automatically and significantly improving recycling efficiency.
- **Mobile Application for On-site Real-Time Monitoring:** Waste management teams can develop a mobile application to track bin status, give notifications, and manage routes to make access and response times smoother.
- **Integrate the system with other smart city systems** such as traffic management and environmental monitoring, thus offering detailed city-wide data insights that can enable coordinated resource management and policy planning.
- **Automated Cleaning and Maintenance Alerts:** It could utilize predictive maintenance algorithms to detect sensor malfunctions or deterioration and automatically send

alerts for calibration, replacement, or cleaning of sensors. This would enhance the reliability of the system and reduce the number of manual checks required.

- **Public Notification and Engagement Systems:** It would integrate with public apps that notify citizens about the availability of bins to be collected to encourage proper waste disposal from the residents' side, which can likely minimize littering in busy areas.
- **Carbon Footprint Monitoring:** The system could monitor the carbon saved through wastes captured in an optimized manner, hence giving statistics on the impact on the environment via minimized usage of fuel and help cities get onto the sustainability pathway.

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APPENDICES

APPENDIX_A

FRONT PAGE OF THE CONFERENCE PAPER

Smart Garbage Monitoring System Using IOT With Weight-Sensing

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ABSTRACT

- The clean city always requires the hard work of individuals working on the ground floor to empty the trash cans whenever they get full. It is not a time-pattern activity to fill up a rubbish bin, and it can fill up very quickly or take a long time to fill up. IOT Garbage Monitoring with Weight Sensing is an innovative step towards the betterment of the efficiency and smoothening of this process. This system monitors the garbage cans and informs the users through a web page about the volume of rubbish collected in the process. It achieves this by the system through ultrasonic sensors placed over the waste bins to measure their depth against the rubbish level. We also have weight sensors attached under the trash cans. Therefore, the system sends the weight and the level of full trash cans over the internet. The advantage of this hybrid sensing is that it makes the authorities know the weight of the rubbish bin that is being lifted. The 12V transformer powers the system. It indicates the quantities and amount of waste that has been collected in the dumpsters on the LCD screen. The web page, though, is designed to show it to the person monitoring it. The rubbish collected will be shown on the internet through color through graphical garbage cans. Besides its level, the weight of each rubbish container shall appear. The LCD panel shows the quantity of waste. The mechanism gives an alarm whenever the quantity of waste collected crosses a predetermined limit. Therefore, this mechanism teaches and keeps the town clean.

Keywords : Internet of Things, Waste Management, ESP32,IR-sensor,dustbin

1. INTRODUCTION

Cities are developing everywhere in the world to host an ever-growing population, leading to waste

management as an integral part of urban infrastructure. Population growth requires collection and disposal management of wastes in more complex ways, especially in heavily populated cities. Over-flown trash cans not only create sanitary problems but also contribute to less attractive and health-conscious public places. [1]Traditional collection methods traditionally work with tight timetables, resulting in inefficient routes, collections of partially empty bins, and missed collections of fully full bins. With routine rounds taken for waste collection, this has resulted in increased fuel consumption, higher operating costs, and a greater carbon footprint. [2]There is a need for a smarter and agile waste management system to solve these inefficiencies and promote cleaner urban surroundings. Paradigm-shifting opportunities abound for the improvement of waste management systems through real-time data collection, monitoring, and automation capabilities offered by the Internet of Things. IoT refers to the process of creating an internet connection between sensors and devices to facilitate uninterrupted data flow and communication. [3] IoT can be employed in waste management wherein the weight and fill level of garbage bins can be monitored. This is helpful to provide useful information, which later may be used to schedule and optimize the routes. Thus, the traditional fill-level monitoring system is fused with additional weight-based data in an IoT-based garbage monitoring system with weight sensing to serve as a more holistic and accurate waste management solution. This dual-sensing technology allows cities to collect only when necessary, hence cutting costs while advocating for environmental sustainability. The following are the primary goals of the IoT Garbage Monitoring System whose objectives are to be mainly carried out by the weight-sensing IoT waste monitoring system : Minimal

APPENDIX_B

We have submitted the Conference paper, till now we did not receive the acceptance letter, and this is the Acknowledgement of Conference paper as we attached it below, Once we receive Acceptance letter we will attach it.

The following submission has been created.

Track Name: ICERCS2024

Paper ID: 1263

Paper Title: Smart Garbage Monitoring System Using IOT With Weight-Sensing

APPENDIX_C

SOURCE CODE

```
#include <WiFi.h>
#include <HTTPClient.h>
#define PIR_PIN 13 // GPIO pin connected to the PIR sensor output
const char* ssid = "hotspot"; // Replace with your Wi-Fi SSID
const char* password = "12345678"; // Replace with your Wi-Fi password
const String telegramToken =
"7806170762:AAGIovvqQtRZlxp9zQ6Ld8FYOP8M1EewFxo"; // Replace with your
Telegram bot token
const String chatId = "5347148803"; // Replace with your chat ID
bool motionSent = false; // Flag to prevent repeated messages
void setup() {
  Serial.begin(115200);
  pinMode(PIR_PIN, INPUT);
  // Connect to Wi-Fi
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Connecting to WiFi...");
  }
  Serial.println("Connected to WiFi");
}
void loop() {
  int motionDetected = digitalRead(PIR_PIN);
  if (motionDetected == LOW && !motionSent) { // Motion detected and message not
sent yet
```

```

    Serial.println("Motion Detected!");
    sendTelegramMessage("The garbage bin is ready to be emptied!!!");
    motionSent = true; // Set flag to avoid repeated messages
} else if (motionDetected == HIGH) {
    Serial.println("No Motion");
    motionSent = false; // Reset flag when no motion
}
delay(1000); // Wait 1 second before the next reading
}

void sendTelegramMessage(String message) {
    if (WiFi.status() == WL_CONNECTED) { // Ensure Wi-Fi is connected
        HTTPClient http;

        String url = "https://api.telegram.org/bot" + telegramToken + "/sendMessage?chat_id="
+ chatId + "&text=" + message;

        http.begin(url);
        int httpResponseCode = http.GET();
        if (httpResponseCode > 0) {
            Serial.print("Message sent, Telegram response code: ");
            Serial.println(httpResponseCode);
        } else {
            Serial.print("Error sending message, code: ");
            Serial.println(httpResponseCode);
        }
        http.end();
    } else {
        Serial.println("WiFi not connected")
    }
}

```

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